Practical - 2

Study of different types of Network cables

Aim:

Study of different types of network cable

a.) Understand different types of Network cables

Different types of cables used in networking are

1.) Unshielded Twisted Pair (UTP) Cable
2.) Shielded Twisted Pair (STP) cable
3.) Coaxial Cable
4.) Fibre optic Cable

Cable type	Category	Maximum data transmission	Adv / disadv	Application/ use	Image
UTP	Category 3 Category 5 Category 5e	10 bps Upto 100 Mbps	Advantages • Cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantage • More prone to EMI and Noise	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
STP	Category 6,6a	10Gbps	Advantages • Shielded • Faster than UTTP • Less susceptible to noise and interference Disadvantages • Expensive • Greater installation effort	Gigabit Ethernet 10G Ethernet (55m) widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet 10G Ethernet (100m)	

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv • High bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadv • Limited distances • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connection	
Fibre optics Cable	Single mode Multimode	100 Gbps	Advan • High speed • High bandwidth • High security • Long distance Disadv • Expensive • Requires high Installer	• Max dis of Fibre optics Cable is around 100	